

ABER: Actuation-Aware Braking-Envelope Recovery for Recursively Safe Robot Learning

Anonymous submission

Abstract

Safe reinforcement learning must constrain actions a robot can execute, while avoiding a shield-induced loss of task completion. We introduce ABER (Actuation-Aware Braking-Envelope Recovery), a sampled-data shield posed directly on forward/turn velocity setpoints. A calibrated stopping envelope gives a closed-form next-speed cap; the certified branch retains the requested turn, while an executable $(0, 0)$ witness preserves recursive feasibility. Under trace-testable servo envelopes, we prove collision avoidance for any finite set of static obstacles and extend the result to measured stopping envelopes and bounded geometry error. Strict recovery is safe but can create long horizon-saturating stalls outside an accepted empirical envelope. We therefore introduce ABER-LR, an explicitly uncertified liveness extension that combines an inflated intervention margin with latched obstacle-relative restoration until certificate re-entry. We evaluate the joint end-point collision-free success rather than safety or success alone. Across 30 independently trained policies spanning two training modes and three Safety-Gymnasium tasks, ABER-LR improves collision-free success over strict ABER in all six mode-task groups and 28/30 checkpoints. The checkpoint-level mean gain is 9.67 points, with collision and timeout reductions of 2.89 and 9.79 points. Relative to unfiltered execution, collision falls by 38.97 points and collision-free success rises by 3.59 points on average, although raw success falls and the gain is not universal across tasks. The results separate the certified safety layer from empirical safety-liveness coordination rather than extending the theorem to restoration steps.

Introduction

Reinforcement-learning policies for mobile robots commonly output a normalized forward/turn command. The low-level controller holds this command for a finite period and tracks a translational-speed and turn-rate setpoint. In contrast, many safety arguments assume a fully actuated Cartesian acceleration. This gap matters most in the fall-back branch: projecting a radial acceleration onto the current heading may neither satisfy the derived constraint nor stop the vehicle.

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We study a sampled state $x_k = (q_k, \theta_k, s_k)$, where $q_k \in \mathbb{R}^2$ is position, θ_k is heading, and $s_k \geq 0$ bounds translational speed. The deployed action is $u_k = (u_k^f, u_k^\omega) \in [-1, 1]^2$. Our objective is not to reintroduce a Cartesian action behind this interface, but to filter the two motor-level setpoints themselves.

Stopping distance has long appeared in adaptive-cruise CBFs (Ames, Grizzle, and Tabuada 2014; Ames et al. 2017), feasibility analysis (Xiao, Belta, and Cassandras 2022), and optimal CBF construction (Beaver 2024). Sampled-data CBFs explicitly account for zero-order hold (Breedon, Garg, and Panagou 2022; Taylor et al. 2022), zero-order CBFs avoid derivative models (Tan et al. 2025), and collision-cone CBFs encode velocity-obstacle geometry (Tayal et al. 2026). Therefore, we do not claim $s^2/(2a_b)$ as new. The unresolved deployment question is narrower: how can that reserve yield an action-space projection and a recovery policy that are both executable by a forward/turn velocity servo, including when several obstacles are simultaneously active?

ABER answers this question with a scalar next-speed cap. For clearance d and sample period Δt , the largest admissible speed z solves

$$\Delta t z + \frac{z^2}{2a_b} \leq d. \quad (1)$$

If the current speed is below the cap, only the forward setpoint is clipped; the requested turn is retained. Otherwise the vehicle receives the zero-speed, zero-turn recovery. This conservative construction is independent of obstacle bearing. That loss of directional permissiveness buys two useful properties: one recovery protects all static obstacles at once, and the online work is $O(m)$ for m obstacles without a QP.

The same conservatism creates a second deployment problem. When the empirical plant leaves the accepted envelope, repeated $(0, 0)$ commands can reduce collision yet occupy the rest of the task horizon. We treat this as an indicator-coordination failure: neither collision nor raw success alone measures a successful safe completion. Our empirical extension ABER-LR couples earlier intervention to a latched obstacle-relative restore, and uses the joint event success $\wedge \neg$ collision as its primary endpoint. Restore is deliberately excluded from the certified witness.

Our contributions are:

1. an actuation-aware safety-filter design that certifies the

executed sampled-data transition rather than an auxiliary Cartesian input, instantiated directly in deployed (forward, turn) space: its closed-form speed cap is maximal within a bearing-independent one-step envelope, is the unique least-change projection for any positive diagonal action metric, and costs $O(m)$ for m obstacles without a QP;

2. an executable $(0, 0)$ recovery witness and a recursive-feasibility and collision-avoidance theorem for any finite set of static obstacles, including explicit closure of the speed bound, a monotone-envelope generalization, and an uncertainty-inflated certificate; and
3. ABER-LR, an explicitly uncertified liveness extension, and a falsification-first evaluation that separates certificate validity, geometric collision, raw success, and collision-free success. The learned study uses 30 saved checkpoints and 15,000 matched filter-off/on pairs across strict ABER and ABER-LR, retaining adverse task and efficiency outcomes.

Related Work

Constraint manifolds and CBFs. ATACOM maps policy actions onto a constraint manifold and establishes continuous-time invariance through null-space dynamics (Liu et al. 2022; Liu 2024; Liu et al. 2025). CBF-QPs similarly modify a nominal command through an online constrained optimization (Ames et al. 2017); HOCBFs treat higher-relative-degree constraints (Xiao and Belta 2022). These are related projection mechanisms, but a guarantee is tied to the dynamics and action map used in its derivation. Our method retains the extended-state manifold view while replacing a Cartesian tangent cone by a setpoint-level sampled-data admissible set.

Sampling and feasibility. Discrete CBFs enforce a one-step condition on a discrete model (Agrawal and Sreenath 2017; Zeng, Zhang, and Sreenath 2021). Breeden, Garg, and Panagou derive sufficient conditions for zero-order-held CBF controllers (Breeden, Garg, and Panagou 2022), while Taylor et al. use approximate discrete-time models (Taylor et al. 2022) and Tan et al. develop ZOCBFs (Tan et al. 2025). These general frameworks can cover unicycle systems when appropriate bounds are available. ABER specializes the construction to bounded braking: the resulting scalar inverse is closed form, and feasibility is witnessed by the exact command sent to the robot.

Velocity-aware geometry. Braking constraints and input compatibility are central in CBF feasibility (Xiao, Belta, and Cassandras 2022; Beaver 2024). Collision-cone CBFs instead use relative position and velocity to exclude collision courses (Tayal et al. 2026). Our certificate is more conservative because it does not credit a favorable turn or obstacle bearing. It consequently extends to multiple static obstacles without requiring intersection feasibility of directional half-spaces.

Safe RL and reachability. Safety layers have been combined with policy optimization (Cheng et al. 2019; Taylor et al. 2020); Lagrangian algorithms enforce expected CMDP costs rather than per-state invariance (Achiam et al. 2017;

Ray, Achiam, and Amodei 2019). Hamilton–Jacobi reachability provides stronger set-based machinery at greater offline cost (Mitchell, Bayen, and Tomlin 2005; Bansal et al. 2017; Fisac et al. 2019). Our optional *sim-BRT* is not an HJ solve: it is a sampled nine-direction forward rollout used only as a reward-shaping term. We therefore report native and shaped training separately.

Problem Formulation

Executable Action Maps

Let a nominal policy π return $u_{\text{RL}} \in [-1, 1]^2$ at every sample. A shield is a map

$$\Phi : (x_k, u_{\text{RL},k}, \mathcal{O}) \mapsto u_k, \quad u_k = (u_k^f, u_k^\omega), \quad (2)$$

where $\mathcal{O}$ is the finite obstacle description and u_k is the exact command held by the low-level velocity servo. Safety must therefore be proved for the transition relation induced by u_k , not for an auxiliary Cartesian input that is never sent to the plant. The policy itself is unrestricted: the certificate must hold for every proposed u_{RL} , including actions far outside the one-step admissible set.

This distinction is structural for nonholonomic platforms. Write $e(\theta) = (\cos \theta, \sin \theta)$ and $e_\perp(\theta) = (-\sin \theta, \cos \theta)$. For the acceleration-level unicycle idealization

$$\dot{q} = se(\theta), \quad \dot{s} = a, \quad \dot{\theta} = \omega, \quad (3)$$

the Cartesian acceleration is $\ddot{q} = ae(\theta) + s\omega e_\perp(\theta)$.

Lemma 1 (Cartesian fallback mismatch). At $s = 0$, the instantaneous image of every bounded unicycle input (a, ω) under (3) is the one-dimensional set $\{ae(\theta)\}$. Hence a Cartesian safety fallback with nonzero component along $e_\perp(\theta)$ has no realizing unicycle input at that state.

Proof. Substituting $s = 0$ into $\ddot{q}$ removes the only lateral term, independently of ω . Therefore $e_\perp(\theta)^\top \ddot{q} = 0$ for every admissible input, which excludes any requested lateral component. $\square$

At $s > 0$, the lateral component remains coupled to $s\omega$ and to the turn bound; it is still not an independent Cartesian degree of freedom. Lemma 1 does not say that Cartesian CBFs cannot be specialized to unicycles. It shows why feasibility in an unconstrained Cartesian action space is insufficient: the proof must either include the platform action map or, as ABER does, be posed directly at the deployed interface.

Recursive Shield Objective

For a state set $\mathcal{C}$, call a shield *recursively feasible* if every state in $\mathcal{C}$ admits an action returned by the shield and every accepted plant transition under that action ends in $\mathcal{C}$. This is stronger than producing a feasible optimization result at the current sample: it requires the returned action to preserve feasibility of the next sample. Our goal is a map Φ satisfying this property for arbitrary nominal policy actions, while changing the turn request only when the certified recovery branch is necessary.

The certificate uses path length rather than a bearing-specific radial velocity. If a realized trajectory over one sample has length ℓ , then for every fixed point o_i ,

$$\|q^+ - o_i\| \geq \|q - o_i\| - \|q^+ - q\| \geq \|q - o_i\| - \ell. \quad (4)$$

This bound is independent of heading and is simultaneously valid for every static obstacle. It is conservative when the robot turns away, but avoids an intersection of directional constraints whose feasibility may depend on an unavailable Cartesian action.

Actuation-Aware Filter

Model and Certificate Interface

Let static obstacle i have center o_i and keepout radius r_i , including robot body and uncertainty margins. Define

$$d_i(q) = \|q - o_i\| - r_i, \quad (5)$$

$$c_i(q, s) = d_i(q) - D(s), \quad D(s) = \frac{s^2}{2a_b},$$

where $a_b > 0$ is a conservative braking capability and D is a validated upper bound on stopping path length. The safe set is $\mathcal{C} = \{(q, s) : c_i(q, s) \geq 0 \forall i\}$.

The theorem uses a plant acceptance interface rather than a fictitious acceleration input. For a held forward/turn setpoint, write s^+ for the next speed and ℓ for path length during the interval. We require:

A1 (normal servo envelope). For a normal action,

$$s^+ \leq z := \max\{s, v_{\max}|u^f|\}, \quad \ell \leq \Delta t z. \quad (6)$$

A2 (recovery envelope). The executable command $u^{\text{rec}} = (0, 0)$ satisfies

$$\ell^{\text{rec}} + D(s^+) \leq D(s). \quad (7)$$

Both conditions are directly testable from command, pose, and speed traces. For the semi-implicit bounded-braking model $s^+ = \max\{s - a_b \Delta t, 0\}$ and $\ell^{\text{rec}} \leq \Delta t s^+$, (7) can be checked exactly. If $s < a_b \Delta t$, both upper bounds are zero. Otherwise,

$$\begin{aligned} \ell^{\text{rec}} + D(s^+) &\leq \Delta t(s - a_b \Delta t) + \frac{(s - a_b \Delta t)^2}{2a_b} \\ &= D(s) - \frac{1}{2}a_b \Delta t^2 \leq D(s). \end{aligned} \quad (8)$$

Other low-level controllers may use a tabulated stopping envelope D ; only its monotonicity and (7) are needed by the proof.

Projection and Recovery

Let $d_{\min} = \min_i d_i(q)$. Inverting (1) gives

$$z_{\max}(d_{\min}) = \min\{v_{\max}, -a_b \Delta t + \sqrt{(a_b \Delta t)^2 + 2a_b d_{\min}}\}, \quad (9)$$

with $z_{\max} = 0$ when $d_{\min} < 0$. Algorithm 1 clips the forward setpoint when the normal envelope can be certified and otherwise executes recovery. Reverse commands are covered through $|u^f|$.

Proposition 1 (sharp cap within the envelope). For $d \geq 0$, let $g(z) = \Delta t z + z^2/(2a_b)$. Equation (9) returns

$$z_{\max}(d) = \max\{z \in [0, v_{\max}] : g(z) \leq d\}.$$

Algorithm 1 ABER forward/turn projection

Require: q, s, u_{RL} , obstacles, $\Delta t, v_{\max}, a_b$

- 1: $d_{\min} \leftarrow \min_i (\|q - o_i\| - r_i)$
- 2: compute $z_{\max}$ by (9)
- 3: **if** $s \leq z_{\max}$ **then**
- 4: $u^f \leftarrow \text{clip}(u_{\text{RL}}^f, -z_{\max}/v_{\max}, z_{\max}/v_{\max})$
- 5: $u^\omega \leftarrow u_{\text{RL}}^\omega$
- 6: **else**
- 7: $(u^f, u^\omega) \leftarrow (0, 0)$ {certified recovery}
- 8: **end if**
- 9: **return** (u^f, u^ω)

If $z_{\max} < v_{\max}$, every $z > z_{\max}$ violates the one-step inequality. Indeed, $g'(z) = \Delta t + z/a_b > 0$ on $z \geq 0$, and (9) is its nonnegative inverse, clipped only at $v_{\max}$. Thus the cap is not merely sufficient but maximal for the bearing-independent A1 bound. This does not claim maximality among bearing-aware safe actions, which may credit a favorable turn.

The corresponding normal-branch action set is the rectangle

$$\mathcal{U}_N(d) = \left[-\frac{z_{\max}(d)}{v_{\max}}, \frac{z_{\max}(d)}{v_{\max}} \right] \times [-1, 1]. \quad (10)$$

It is used only when $s \leq z_{\max}$; then every $u \in \mathcal{U}_N$ satisfies $\max\{s, v_{\max}|u^f|\} \leq z_{\max}$.

Proposition 2 (least-change interface projection). Let $W = \text{diag}(w_f, w_\omega)$ with $w_f, w_\omega > 0$. In the normal branch, Algorithm 1 is the unique solution of

$$\min_{u \in \mathcal{U}_N(d_{\min})} (u - u_{\text{RL}})^\top W (u - u_{\text{RL}}). \quad (11)$$

Consequently it is the least-change action for any positive diagonal weighting of forward and turn commands within the bearing-independent certificate.

Proof. The objective is strictly convex and the feasible set is a closed convex rectangle. The problem separates by coordinate. Since the turn interval is the original $[-1, 1]$ and u_{RL}^ω already lies in it, its unique minimizer is unchanged. The forward minimizer is the scalar clipping of u_{RL}^f to the first interval in (10), exactly as in Algorithm 1. $\square$

Theorem 1 (recursive feasibility and safety). Let a forward/turn velocity-servo plant satisfy A1–A2, let all obstacles be static, and let $s_0 \leq v_{\max}$. If $(q_0, s_0) \in \mathcal{C}$, then Algorithm 1 is recursively feasible, $(q_k, s_k) \in \mathcal{C}$ for every k , $s_k \leq v_{\max}$, and no keepout is entered. The result holds for any finite number of obstacles.

Proof. Assume $(q_k, s_k) \in \mathcal{C}$ and abbreviate $d_{\min} = d_{\min}(q_k)$. If $s_k \leq z_{\max}$, projection gives $v_{\max}|u_k^f| \leq z_{\max}$, hence A1 gives $s_{k+1} \leq z_{\max}$ and $\ell_k \leq \Delta t z_{\max}$. Distance is 1-Lipschitz along path length, so for every obstacle

$$c_i(q_{k+1}, s_{k+1}) \geq d_i(q_k) - \ell_k - D(s_{k+1}) \quad (12)$$

$$\geq d_{\min} - \Delta t z_{\max} - D(z_{\max}) \geq 0, \quad (13)$$

where the last inequality is the definition of the cap.

If $s_k > z_{\max}$, the algorithm issues $(0, 0)$. A2 yields, for every i ,

$$\begin{aligned} c_i(q_{k+1}, s_{k+1}) &\geq d_i(q_k) - \ell_k^{\text{rec}} - D(s_{k+1}) \\ &\geq d_i(q_k) - D(s_k) = c_i(q_k, s_k) \geq 0. \end{aligned} \quad (14)$$

Thus both branches preserve the complete intersection, and recovery is a feasible witness even when directional constraints would conflict. Finally, A1 preserves $s \leq v_{\max}$ in the normal branch; braking cannot increase it in recovery. Induction completes the proof. $\square$

General Envelopes and Robust Bounds

The quadratic $D(s) = s^2/(2a_b)$ gives the closed form in (9), but the invariance argument is not tied to constant deceleration. Let $D : [0, v_{\max}] \rightarrow \mathbb{R}_{\geq 0}$ be continuous and nondecreasing with $D(0) = 0$, and define

$$\begin{aligned} G_D(z) &= \Delta t z + D(z), \\ z_D(d) &= \max\{z \in [0, v_{\max}] : G_D(z) \leq d\}. \end{aligned} \quad (15)$$

For an empty set take $z_D(d) = 0$ and enter recovery. A measured piecewise linear or tabulated stopping envelope is therefore admissible; when G_D is strictly increasing, z_D is unique and can be obtained by a scalar monotone search. The evaluated ABER implementation retains the quadratic closed form, so this extension is a scope result rather than a runtime claim for lookup tables.

Theorem 2 (monotone-envelope invariance). Replace (9) in Algorithm 1 by $z_D(d_{\min})$. Suppose the normal transition obeys A1 and recovery obeys $\ell^{\text{rec}} + D(s^+) \leq D(s)$ for this D . If $d_i(q_0) - D(s_0) \geq 0$ for every static obstacle, then the generalized filter is recursively feasible and preserves $d_i(q_k) - D(s_k) \geq 0$ for all i and k .

Proof. In the normal branch, $s^+ \leq z_D$ and $\ell \leq \Delta t z_D$. Monotonicity of D and (4) give

$$\begin{aligned} d_i(q^+) - D(s^+) &\geq d_i(q) - \Delta t z_D - D(z_D) \\ &\geq d_{\min} - G_D(z_D) \geq 0. \end{aligned} \quad (16)$$

In recovery, (4) and the accepted recovery inequality give

$$d_i(q^+) - D(s^+) \geq d_i(q) - D(s) \geq 0. \quad (17)$$

Thus either branch preserves the entire obstacle intersection. Recovery is always an available witness on the certified set, and induction proves the claim. $\square$

The next result turns bounded perception error into the lower-clearance and upper-speed quantities needed by Theorem 2. Let $\hat{q}$, $\hat{o}_i$, $\hat{r}_i$ be reported robot position, obstacle center, and keepout radius, and assume

$$\|q - \hat{q}\| \leq \epsilon_q, \quad \|o_i - \hat{o}_i\| \leq \epsilon_{o,i}, \quad r_i \leq \hat{r}_i + \epsilon_{r,i}. \quad (18)$$

The triangle inequality yields the computable lower bound

$$\underline{d}_i = \|\hat{q} - \hat{o}_i\| - \hat{r}_i - \epsilon_q - \epsilon_{o,i} - \epsilon_{r,i} \leq d_i(q). \quad (19)$$

Corollary 1 (uncertainty-inflated certificate). Let $\bar{s} \geq s$ be a maintained speed upper bound. Suppose A1–A2 are validated for the bound-state transition $\bar{s} \mapsto \bar{s}^+$, and run the generalized filter with $\min_i \underline{d}_i$ and $\bar{s}$. If $\underline{d}_i(q_0) - D(\bar{s}_0) \geq 0$ for all i , then the true clearances satisfy $d_i(q_k) - D(s_k) \geq 0$ for all i, k .

Proof. Equations (18)–(19) give $d_i \geq \underline{d}_i$, while monotonicity gives $D(s) \leq D(\bar{s})$. Theorem 2 preserves the stronger surrogate inequalities $\underline{d}_i - D(\bar{s}) \geq 0$, which imply the claimed true-state inequalities. $\square$

This corollary separates statistical estimation from the deterministic safety argument: confidence bounds or engineering tolerances may supply the ϵ values, but the theorem is conditional on those bounds being valid. Likewise, command latency can be included by validating A1 with an effective hold period Δt_{eff} ; unmodeled delay is not covered.

Necessity and Residual Robustness

The two terms in $G_D(z) = \Delta t z + D(z)$ play different roles. The reserve $D(z)$ pays for stopping after the next sample; $\Delta t z$ pays for motion during the sample itself. The following construction shows that the latter cannot be dropped while retaining a worst-case A1 certificate.

Proposition 3 (necessity of sampled displacement). Let D be strictly increasing, choose $s \in (0, v_{\max})$, and set $d = \Delta t s + D(s)$. Assume the root of $D(z) = d$ lies below $v_{\max}$. A cap using only $D(z) \leq d$ admits $\tilde{z} = D^{-1}(d) > s$. There exists an A1 transition from this certified state whose next certificate is negative.

Proof. The initial certificate is $d - D(s) = \Delta t s > 0$. Select an obstacle directly along the realized path and an A1-admissible transition attaining $s^+ = \tilde{z}$ and $\ell = \Delta t \tilde{z}$. Because $D(\tilde{z}) = d$, its next margin is

$$d - \ell - D(s^+) = d - \Delta t \tilde{z} - D(\tilde{z}) = -\Delta t \tilde{z} < 0. \quad (20)$$

Thus the truncated cap can cross the certificate before penetrating the keepout. In contrast, the full cap has $G_D(s) = d$ and therefore does not admit a larger speed. $\square$

The recovery branch is also a semantic part of the certificate, rather than an arbitrary low-speed replacement. A1 bounds normal commands but does not require them to begin braking immediately.

Proposition 4 (a nonzero fallback is not certified by A1). For every $s \in (0, v_{\max}]$ and boundary state $d = D(s)$, there is a velocity-servo transition relation satisfying A1 for nonzero commands and A2 for $(0, 0)$ such that every nonzero fallback leaves the certified set in one sample.

Proof. Consider a one-sample setpoint lag: after any nonzero fallback, the plant keeps $s^+ = s$, travels $\ell = \Delta t s$ toward the obstacle, and only then accepts the new setpoint. This satisfies A1 because $s \leq \max\{s, v_{\max}|u^f|\}$ and $\ell \leq \Delta t \max\{s, v_{\max}|u^f|\}$. Its next margin is

$$d - \ell - D(s^+) = D(s) - \Delta t s - D(s) = -\Delta t s < 0. \quad (21)$$

Define the response to the exact command $(0, 0)$ instead by any braking transition satisfying A2, such as (8). That command preserves the boundary by Theorem 1, while the nonzero fallback does not. $\square$

Propositions 3–4 are worst-case statements about the accepted interface, not claims that every physical servo attains the adversarial transitions. They explain why the component study tests certificate margin at the constructed boundaries $d = \Delta t s + D(s)$ and $d = D(s)$, rather than relying only on observed collision. They also delimit the claim: a nonzero recovery could be used if its own trace envelope replaced A2 and were proved to preserve the reserve.

Trace calibration may leave bounded normal-branch residuals. Suppose a validation procedure establishes, for constants $\varepsilon_s, \varepsilon_\ell \geq 0$,

$$s^+ \leq s + \varepsilon_s, \quad \ell \leq \Delta t s + \varepsilon_\ell, \quad (22)$$

where $z = \max\{s, v_{\max}|u^f|\}$ and D is defined through $v_{\max} + \varepsilon_s$. Define the tightened cap

$$z_{D,\varepsilon}(d) = \max\{z \in [0, v_{\max}] : \Delta tz + \varepsilon_\ell + D(z + \varepsilon_s) \leq d\}. \quad (23)$$

If the set is empty, ABER enters recovery.

Proposition 5 (residual-robust normal branch). Under (22), replacing z_D by $z_{D,\varepsilon}$ makes every accepted normal transition preserve all obstacle certificates. If the unchanged recovery branch satisfies A2, the recursive conclusion of Theorem 2 follows.

Proof. For every obstacle, monotonicity of D , (4), and the tightened cap give

$$\begin{aligned} d_i(q^+) - D(s^+) &\geq d_i(q) - \Delta tz - \varepsilon_\ell - D(z + \varepsilon_s) \\ &\geq d_{\min} - \Delta tz - \varepsilon_\ell - D(z + \varepsilon_s) \geq 0. \end{aligned} \quad (24)$$

The recovery step is unchanged, so the same induction applies. $\square$

For the quadratic reserve, (23) remains a scalar quadratic inequality and therefore retains a closed-form positive root. Setting both residuals to zero recovers (9). Additive errors are not silently absorbed into an empirical claim: they must be predeclared from trace bounds, and a platform whose residuals exceed them is outside the certificate.

With lower clearances already computed, finding their minimum takes $O(m)$ time and $O(1)$ streaming memory, while all remaining quadratic-envelope operations are constant time. Hence the multi-obstacle guarantee does not conceal an online optimization problem.

Regularity and Obstacle Composition

Besides being closed form, the quadratic cap has useful regularity. Before the $v_{\max}$ saturation, implicit differentiation of $\Delta tz + D(z) = d$ gives

$$\frac{dz_{\max}}{dd} = \frac{a_b}{\sqrt{(a_b \Delta t)^2 + 2a_b d}} = \frac{1}{\Delta t + z_{\max}/a_b}, \quad (25)$$

and

$$\frac{d^2 z_{\max}}{dd^2} = -\frac{a_b^2}{((a_b \Delta t)^2 + 2a_b d)^{3/2}} < 0. \quad (26)$$

Thus $z_{\max}$ is nondecreasing and concave in clearance. Since the first derivative is at most $1/\Delta t$, saturation at $v_{\max}$ preserves the global bound

$$|z_{\max}(d_1) - z_{\max}(d_2)| \leq \frac{|d_1 - d_2|}{\Delta t}, \quad d_1, d_2 \geq 0. \quad (27)$$

This bound is descriptive of the cap, not a substitute for the one-sided uncertainty treatment in Corollary 1: using a noisy clearance without a valid lower bound can still overestimate safety.

Proposition 6 (normal-projection sensitivity). Let $P_d(a) = \text{clip}(a, -z_{\max}(d)/v_{\max}, z_{\max}(d)/v_{\max})$. For any $a_1, a_2 \in [-1, 1]$ and $d_1, d_2 \geq 0$,

$$|P_{d_1}(a_1) - P_{d_2}(a_2)| \leq |a_1 - a_2| + \frac{|d_1 - d_2|}{v_{\max} \Delta t}. \quad (28)$$

Proof. For fixed d , scalar projection onto a closed interval is nonexpansive, so $|P_d(a_1) - P_d(a_2)| \leq |a_1 - a_2|$. For fixed a , changing the symmetric interval radius changes the clipped value by no more than the change in that radius. Apply the triangle inequality and (27). $\square$

Therefore the normal-branch forward action is continuous in both nominal forward command and certified clearance, and the turn action is exactly the nominal turn. The complete shield need not be continuous at $s = z_{\max}(d)$ because crossing that surface activates the discrete recovery $(0, 0)$. This is an intentional safety switch; the theorem does not claim smoothness or absence of chattering at that boundary.

For multiple obstacles, the scalar reduction itself adds no conservatism beyond the bearing-independent path envelope. Indeed, because G_D is common to all obstacles,

$$[G_D(z) \leq d_i \ \forall i] \iff G_D(z) \leq \min_i d_i. \quad (29)$$

Proposition 7 (monotone obstacle composition). For a finite obstacle set $\mathcal{O}$, the generalized cap satisfies

$$z_D\left(\min_{i \in \mathcal{O}} d_i\right) = \min_{i \in \mathcal{O}} z_D(d_i). \quad (30)$$

If $\mathcal{O} \subseteq \mathcal{O}'$, then the cap for $\mathcal{O}'$ cannot exceed that for $\mathcal{O}$. Hence adding an obstacle cannot enlarge the certified normal-action set, while the same recovery witness continues to preserve the complete intersection.

Proof. Equation (29) shows that the feasible scalar-speed set for all obstacles is exactly the intersection of their individual feasible intervals. Each interval begins at zero, so the endpoint of the intersection is the minimum of the individual endpoints. Set inclusion gives the monotonicity claim, and Theorem 2 gives recovery feasibility. $\square$

Proposition 8 (inter-sample keepout avoidance). Under the assumptions of Theorem 1, the certificate prevents keepout entry throughout each held control interval, not only at its endpoints.

Proof. Let $q(\tau)$ be any intermediate configuration and let $\ell(\tau)$ be the path length accumulated since q_k . The distance-to-set map is 1-Lipschitz, so $d_i(q(\tau)) \geq d_i(q_k) - \ell(\tau)$ for every obstacle. On the normal branch, $\ell(\tau) \leq \ell_k \leq \Delta t z_{\max}$ and $d_i(q_k) \geq \Delta t z_{\max} + D(z_{\max})$; hence $d_i(q(\tau)) \geq D(z_{\max}) \geq 0$. On the recovery branch, $\ell(\tau) \leq \ell_k^{\text{rec}}$ and A2 gives $\ell_k^{\text{rec}} + D(s_{k+1}) \leq D(s_k)$. Since $d_i(q_k) \geq D(s_k)$, it follows that $d_i(q(\tau)) \geq D(s_{k+1}) \geq 0$. Thus neither branch can cross a declared static keepout between samples. $\square$

Liveness-Aware Empirical Recovery

Strict ABER uses $(0, 0)$ because A2 makes it a common witness for every static obstacle. In Safety-Gym, however, an uncertified robot can approach zero speed without re-entering $\mathcal{C}$; repeating the witness then creates a recovery tail rather than task progress. ABER-LR addresses this empirical failure without modifying Algorithm 1 or Theorem 1.

Let j index the minimum-clearance obstacle and let $n = (q - o_j)/\|q - o_j\|$ point away from it. Resolving n in the body frame gives $n_f = e(\theta)^\top n$ and $n_\perp = e_\perp(\theta)^\top n$. When the inflated certificate is negative and $s \leq s_{\text{stop}}$, ABER-LR starts the bounded restore

$$u^{\text{LR}} = (\text{clip}(\alpha n_f, -\alpha, \alpha), \text{clip}(\kappa n_\perp, -1, 1)), \quad (31)$$

Navigation condition	Epis.	Viol.	Coll.	Succ.
Full ABER, $\rho \geq 1$	1200	0	0	1186
No sampled term, $\rho \geq 1$	1200	1200	481	719
No recovery, $\rho \geq 1$	1200	1002	298	902
Full ABER, $\rho < 1$	1200	873	539	660

Table 1: Component navigation study. “Viol.” counts episodes that leave the braking-envelope certificate; “Coll.” counts keepout entry. The 14 calibrated full-ABER nonsuccesses are horizon timeouts, not collisions.

with a small nonzero translation floor to avoid yaw-only stalling. The restore is latched until certificate re-entry; without the latch, one restore step raises speed and the next step returns to hard braking. Experiments fix $s_{\text{stop}} = 0.05$, $\alpha = 0.25$, $\kappa = 1$, $a_b = 0.9$, and an inflated 0.10-m margin before cross-task evaluation.

Equation (31) is an empirical outer recovery controller, not an A2 witness. Its steps are marked uncertified, and every theorem and certificate count below refers only to strict ABER under the accepted plant interface. This separation lets us test whether liveness can be recovered without retroactively labeling the fallback as safe.

Scope and deployment. A1–A2 are trace-testable plant contracts, not consequences of sending a low setpoint. Deployment requires measuring the effective hold period, fitting a conservative D , validating normal and recovery traces, and inflating geometry as in Corollary 1. Moving obstacles, unmatched delay, starts outside $\mathcal{C}$, or a platform that fails either trace test are not covered.

Experiments

We test component necessity, safety and task completion on the accepted interface, learned-policy behavior outside a globally accepted envelope, and reproducibility. Protocols, seeds, metrics, and failures are fixed before aggregation; certificate violations, geometric collisions, and Safety-Gym costs are reported separately because they are not interchangeable.

Falsification-First Component Study

We first test the two algorithmic branches and the calibration premise rather than infer them from a collision count. The grid crosses $\Delta t \in \{0.02, 0.05, 0.10, 0.20\}$ s, actual-to-configured braking ratio $\rho \in \{0.5, 0.75, 1.0, 1.25\}$, and $m \in \{1, 4, 8\}$ obstacles. We compare full ABER with (i) a cap that omits $\Delta t z$, and (ii) a variant that keeps the cap but clips to a nonzero setpoint instead of issuing $(0, 0)$ when $s > z_{\text{max}}$. Every cell uses 1,000 one-step samples for each of two predeclared boundary challenges and 50 seeded navigation episodes.

The sampled-term challenge starts at $d = \Delta t s + D(s)$, where full ABER is exactly feasible but the shortened cap can over-accelerate. The recovery challenge starts at $d = D(s)$, where A2 certifies $(0, 0)$ but does not certify a nonzero lower setpoint. These constructed trials test the recursive certificate before physical penetration; collision is a secondary end-point.

Scene	Method	Coll. (%)	Succ. (%)	Time (s)	min clr. worst (m)
Single	Braking CBF	0	100	5.80	0.290
	SD-CBF/ZOCBF	0	100	4.90	0.185
	Collision cone	0	100	5.91	0.083
	ABER	0	100	6.46	0.016
Multi	Braking CBF	0	100	7.22	0.220
	SD-CBF/ZOCBF	0	100	5.55	0.046
	Collision cone	0	100	7.33	0.001
	ABER	0	100	7.22	0.016

Table 2: Fixed-config certificate-envelope results (100 episodes per cell). All methods are safe here; only ABER is evaluated against Theorem 1 and has zero certificate violations in both scenes. Minimum clearance is relative to the declared keepout.

In the calibrated boundary controls, full ABER has 0/24,000 violations on each challenge. Omitting the sampled term causes 24,000/24,000 violations on its challenge; omitting recovery does the same on the recovery challenge. With $\rho < 1$, even full ABER violates 24,000/24,000 recovery challenges, as expected when A2 is false. Table 1 shows that the same distinctions persist in navigation: both ablations sometimes make more short-term progress, but only by leaving the certified set.

The complete artifact contains 288,000 boundary records and 7,200 navigation records. All 48 predeclared grid checks pass, and two formal runs produce the same raw SHA-256. Thus the experiment supports necessity within the stated A1–A2 interface; it does not claim that every velocity servo exhibits the particular one-sample nonzero-setpoint delay used to expose the distinction.

Certified-Envelope Comparison

We first evaluate exactly the plant used by Theorem 1. Parameters are $\Delta t = 0.1$ s, $v_{\text{max}} = 1$ m/s, $a_b = a_{\text{acc}} = 0.9$ m/s², $|\omega| \leq 1.5$ rad/s, and $r_i = 0.25$ m. A shared reactive navigator produces the nominal forward/turn action. We sample 100 single-obstacle and 100 four-obstacle layouts from fixed seeds, all starting at rest in $\mathcal{C}$. Baselines are fixed geometry-specialized implementations of a continuous braking-distance CBF, a one-step sampled-data CBF/ZOCBF, and a collision-cone CBF; this benchmark does not claim to reproduce every tuning choice in their source papers. Raw per-episode results and the complete manifest are included with the submission.

Table 2 does not support a claim that other filters are unsafe. It shows instead that ABER’s formal guarantee is obtained without stopping the task: all ABER trials reach the goal. It is slower in the single obstacle case and essentially matches braking CBF time in the multi-obstacle case. Its smaller clearance is expected because the cap uses available stopping reserve rather than a directional stand-off.

Learned-Policy Studies

We next use Safety-Gymnasium Point Goal1, Push1, and Goal2 (Ji et al. 2023). Source inspection gives an environment period of $10 \times 0.002 = 0.02$ s. Twenty zero-setpoint

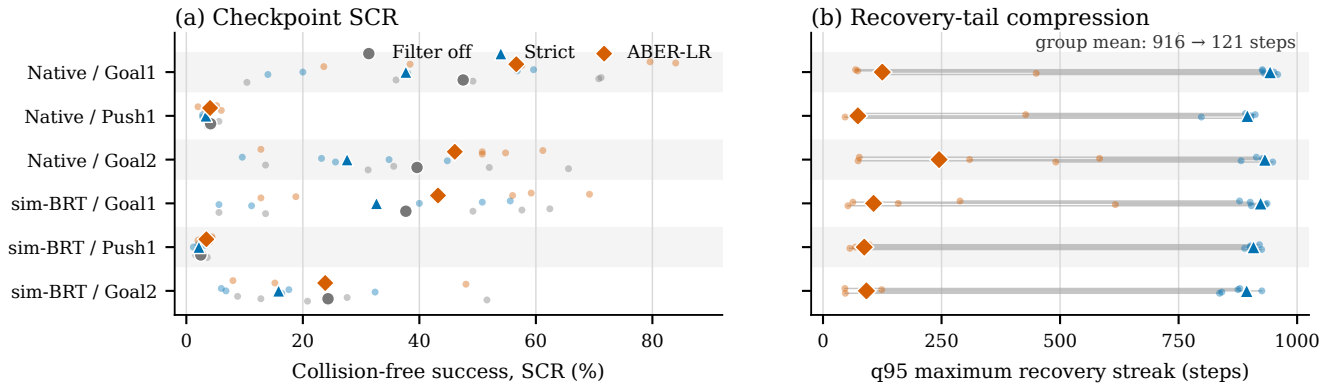


Figure 1: Frozen 30-checkpoint study. Small symbols are checkpoints and large symbols are group means. (a) Collision-free success. (b) Same-checkpoint strict-to-LR q95 recovery-tail contraction (gray links).

braking traces measured a peak speed of 1.468 m/s and a minimum stopping-path equivalent deceleration of 2.06 m/s^2 ; we use the conservative envelope $v_{\max} = 1.5 \text{ m/s}$ and $a_b = 0.9 \text{ m/s}^2$. These are empirical plant acceptance measurements, not evidence that MuJoCo satisfies A1–A2 globally.

Each setting uses PPO for 200K steps and five seeds. Native training is compared with sim-BRT reward shaping, a five-step forward rollout rather than HJ reachability. The supplement retains the unpaired integration matrix; the intervention claim uses the matched execution toggle below.

Paired safety–liveness isolation. We save all 30 policies in the two-mode, three-task, five-seed matrix. Each checkpoint is evaluated with deterministic actions on 250 pre-declared reset seeds under filter-off, strict ABER, and the parameter-frozen ABER-LR. Strict and LR runs use identical checkpoints and initial snapshots; their independently re-run filter-off records agree exactly. This yields 7,500 matched pairs per filtered variant. The generalization unit is the trained checkpoint; episode pairing controls reset variation, and confidence intervals resample checkpoints.

Raw success can count an episode that also collides. We therefore define safe completion as

$$\text{SCR} = \Pr(\text{success} \wedge \neg \text{geometric collision}), \quad (32)$$

and report it with both marginals, unsafe success, and timeout. This joint event introduces no tuned scalarization weight.

Negative tests are classified by the failed gate: collision (safety), success or recovery tail (liveness), SCR (coordination), or consistency across seeds and tasks (transfer). The supplement retains every rejected recovery pilot.

Relative to strict ABER, ABER-LR increases checkpoint-level SCR by 9.67 points (checkpoint bootstrap 95% CI [6.37, 13.33]), success by 9.79 points, and reduces collision and timeout by 2.89 and 9.79 points. All six mode–task means and 28/30 checkpoints improve SCR (Fig. 1). A gate frozen before cross-task transfer—success gain at least 10 points, no collision increase, SCR gain at least 5 points, and SCR improvement in at least four seeds—passes for Native Goal1, Native Goal2, and sim-BRT Goal1. sim-BRT Goal2 improves SCR in 5/5 seeds but misses the 10-point success

Execution	Succ.	Coll.	SCR	Unsafe succ.	Timeout
Filter off	42.48	43.19	25.96	16.52	57.52
Strict ABER	19.92	7.11	19.88	0.04	80.08
ABER-LR	29.71	4.21	29.55	0.16	70.29

Table 3: Equal-weight 30-checkpoint learned-policy summary (%). SCR is $\Pr(\text{success} \wedge \neg \text{geometric collision})$; unsafe success is the complementary successful-but-colliding event.

threshold; both Push1 policies have only 4.6–5.1% unfiltered success, limiting the absolute completion gain.

Relative to filter-off execution, ABER-LR lowers collision by 38.97 points and unsafe success by 16.36 points. Mean SCR rises by 3.59 points ([1.03, 6.37]), but raw success falls by 12.77 points and timeout rises by the same amount. Moreover, SCR improves in only four of six group means and 17/30 checkpoints. Thus the evidence supports indicator coordination on average, not universal dominance over the unfiltered policy.

The mechanism matches the hypothesis: mean recovery occupancy falls from 51.15% to 22.76%, the group-average q95 streak from 916 to 121 steps, and timeout by 9.79 points. In an earlier 250-pair pilot, yaw-only escape left q95 at 958 and gained only 1.6 success points; $a_b = 1.5/2.0$ left q95 above 960 and reduced success. Exit authority, not only recovery entry, is the difficulty. This is liveness evidence, not a restore certificate.

Limitations and Conclusion

Strict ABER assumes an accepted static-obstacle envelope and certified start; Safety-Gym and LR restore remain empirical, and Push1 contacts lie outside the theorem. ABER certifies the transition caused by the executed action; ABER-LR coordinates that conservative witness with liveness, improving collision-free completion over strict ABER in all six groups without extending the theorem to restoration. A directional-restore certificate under measured servo dynamics remains open; LR evidence is empirical. Rejected variants therefore define the mechanism boundary, rather than hidden tuning runs.

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